Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
C		2525966

Analysis of the Olympic Medal Rankings

Abstract

The number of MEDALS of the Olympic Games is an important standard to measure the level of a country's sports. Based on the Olympic medal winning data of past years, this paper uses multiple models to analyze the predicted value of Olympic MEDALS, takes multiple factors into consideration, uses multiple codes to improve the accuracy of the predicted value.Therefore, we can predict national MEDALS to provide suggestions for improving national sports level.

For Task 1:This task is aimed at predicting the results of the medal table at the 2028 Summer Olympics in Los Angeles, USA and the change of the countries' performance.To achieve the goal, we use a linear regression model, which adopts the number of historic medal to be the characteristic factor. We firstly classify all countries based on historical data, selected the countries with relatively stable data, and processed them with the help of the ARIMA model to ensure the accuracy of the prediction. Subsequently, we use the ADF test to determine whether the model met the stability requirements. Finally, we successfully predict the result and analyze which countries might progress or regress by using differential method.We also evaluat the uncertainty, accuracy estimation, and performance of the model.

For Task 2:This task mainly aims to predict which country that has not yet won a medal is most likely to win the first gold medal. We adopt the logistic regression model to analyze the time series and obtained the corresponding results.

For Task 3:This task is aimed to predict the relationship between the sport event and the number of MEDALS won by each country.We use the Apriori algorithm to work it out.

For Task 4: This task is aimed to predict the effect of a great coach on the probability of winning Olympic medals in each country. We use logical analysis to solve this task with typical country cases.

For Task 5:This task mainly aims to illustrate the significance of our model for the Olympic medal table and the value for countries which want to join in the Olmpics. Based on the data and logical analysis, we give several suggestions.

Keywords: Linear Regression Model ARIMA Model Apriori Algorithm

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1.Introduce

1.1Background

The Olympic Game is the most prestigious sporting event in the world. In Olympic Games, each country sends its top athletes to compete for Olympic MEDALS. The Olympic medal table, serving as an important reflection of national comprehensive sports strength ， has been getting a lot of attention. To a certain extent, the Olympic medal table follows the law of the prediction model. Therefore, we can predict the results of the medal table of the upcoming 2028 Los Angeles Olympic Games based on historical data, and provide suggestions for countries to prepare for the Games based on the predicted results to help countries achieve better results in the Olympic Games.

1.2 Restatement of the problem

Task 1 is aimed at predicting the results of the medal table at the 2028 Summer Olympics in Los Angeles, USA and the change of the countries' performance.Based on the research of task1 ， task 2 mainly aims to predict which country that has not yet won a medal is most likely to win the first gold medal.Task 3 is aimed to predict the relationship between the sport event and the number of MEDALS won by each country.Task 4 is aimed to predict the effect of a great coach on the probability of winning Olympic medals in each country.Task 5 mainly aims to illustrate the significance of our model for the Olympic medal table and the value for countries which want to join in the Olmpics.

1.3 Our work

According to the background and historic data, our work includes the following aspects：

1. Conduct data preprocessing to obtain the independent variables required for subsequent models. Simulate the national commercial power values of participating countries based on the linear model and ARIMA model to identify target countries for analysis.

2. Analyze the medal - winning trends of various countries to obtain the final results.

3. Analyze from the dual perspectives of countries and sports events. Use the Apriori algorithm to obtain the final results.

4. Collect relevant materials, identify target countries affected by the famous - coach effect, and conduct analysis in combination with multiple influencing factors.

5. Combine the models with the international sports situation to put forward pertinent suggestions.

2. Assumption

1.The historical medal data follows a certain linear relationship.

2.Historical MEDALS are the main reflection of the national sports level

3.The performance of athletes from all countries is stable

3. Notation

M: The actual number of MEDALS

$\beta_0 \cdots \beta_n$:regression coefficient

$N_0 \cdots N_n$:characteristic factors

M_t: Number of MEDALS

p:The order of the autoregressive term in the ARIMA model

q:The order of the moving average term in the ARIMA model

$M_{i\,2028}$ The predicted number of medals for Country I in 2028

$M_{i\,2024}$ The actual number of medals for Country I in 2024

c_jThe probability of athlete j, who is under significant attention, winning an award.

c_nThe probability of athlete n, who is under significant attention, winning an award.

4. Models

4.1 Analysis of Task 1

This question is to predict the outcome of the 2028 Summer Olympics in Los Angeles, USA. We adopted the ARIMA model to predict the number of MEDALS. The uncertainty, accuracy estimation and performance of the model are evaluated to accurately predict the 2028 Olympic medal table and provide prediction intervals with analyzing which countries are likely to advance or retreat.

4.2 Introduction of the model

Our team systematically processed and analyzed the predictive data using the time series model under the framework of linear regression analysis.

The fundamental formula of linear regression:

$$M = \beta_0 + \beta_1 N_1 + \beta_2 N_2 + \cdots \beta_n N_n + \varepsilon$$

M The actual number of MEDALS

$\beta_0 \cdots \beta_n$:regression coefficient

$N_0 \cdots N_n$:characteristic factors

(Such factors as historical medals and whether it is the host country that affect the number of medals.)

ε: error term

ARIMA Model (Autoregressive Integrated Moving Average Model)

The fundamental formula:

$$M_t = \alpha + \sum_{i=1}^{p} \gamma_i M_{t-i} + \sum_{j=1}^{q} \theta_i \varepsilon_{t-j} + \varepsilon_t$$

M_t: The number of medals

α : constant term

p: The order of the autoregressive (AR) term, indicating the number of periods of historical values used for prediction.

q: The order of the moving average (MA) term, which is used to represent the linear combination of errors.

γ_i、θ_j :autocorrelation

ε_t :white noise

The ARIMA model can obtain the optimal orders p and q through the analysis of the autocorrelation plot and the partial autocorrelation plot. The ARIMA model can also capture random fluctuations, thereby reducing errors and providing more accurate predictions.

4.3 Calculation of Quotient Weight Value

	Rank		Gold			Total				
NOC	mean	std	mean	std	co	mean	std	co	total2024	OFF
Chinese Taipei	42.2	7.362064928	1.4	0.547722558	0.391230398	5.6	4.037325848	0.720951044	7	-1.4
South Africa	43.4	20.29285589	1.6	1.516575089	0.947859431	5.2	3.420526275	0.657793514	6	-0.8
Uzbekistan	41	27.26719641	3	3.31662479	1.105541597	7.6	4.979959839	0.655257874	13	-5.4
Brazil	18	5.14781507	4.6	2.19089023	0.476280485	18.8	1.788854382	0.095151829	20	-1.2
United States	1.2	0.447213595	41.8	5.019960159	0.120094741	115.2	8.526429499	0.074014145	126	-10.8
Spain	16.4	3.361547263	4.8	1.483239697	0.30900827	18.2	1.303840481	0.071639587	18	0.2

4.4 Process

In order to solve the problem, we follow the following steps

4.4.1Processing of stationary data

1. Data Collection

Firstly, we process the collected medal data from previous sessions. Organize the number of gold medals, silver medals, bronze medals, and the total number of medals each country has won in different years to ensure data integrity.

2. Data Processing

Classify the countries:

a. Countries with no recent competition records.

b. Countries that have participated in multiple competitions with continuous and stable performance.

c. Countries that have participated in multiple competitions with unstable performance.

d. Countries that have only started participating in recent years.

e. Countries that have never won an award.

3. Modeling Process

Since the data of the time - series model must meet the stationarity condition, which means the expectation and variance do not change significantly. Among the above five groups, we choose Group b with the highest stationarity for exploration.

Use the ADP test to examine the model: Divide the variance calculated previously by the average number of predicted medals, and then sort the calculated ratios from high to low. Select the three with the smallest ratios for individual chart analysis to prove the accuracy of the model prediction.

4. Prediction Results

We use the trained ARIMA model to predict the number of medals in the 2028 Olympic Games.

The medal table is as follows：

	No	Total	Gold
NOC	mean	mean	mean
United States	1	115.2	41.8
China	2	88.4	38.2
Great Britain	3	62.4	22.2
France	4	43.4	10.8
Australia	5	41.8	13
Japan	6	41.4	15
Germany	7	39.4	13.2
Italy	8	32.6	9.2
South Korea	9	27.2	10.8
Netherlands	10	25	9.2
Canada	11	22.2	5
Brazil	12	18.8	4.6
Spain	13	18.2	4.8
Ukraine	14	16.6	3.6
Hungary	15	16.4	6.2
New Zealand	16	16	6
Cuba	17	16	4.4
Kenya	18	12.6	4.4
Poland	19	11.4	2.8
Kazakhstan	20	10.4	1.6
Denmark	21	10.2	2.2
Jamaica	22	10	4
Azerbaijan	23	9.4	1.2
Czech Republic	24	8.8	3
Sweden	25	8.8	2
Iran	26	8.4	3.4
Switzerland	27	7.8	2.2
Uzbekistan	28	7.6	3
Turkey	29	7.4	1
Croatia	30	7.2	2.6
Georgia	31	7	2.2
Norway	32	6.6	2.6
Romania	33	6.6	2.2
Ethiopia	34	6.2	2
Serbia	35	5.8	1.8
Colomba	36	5.8	0.8
Belgium	37	5.6	2
Chinese Taipei	38	5.6	1.4
South Africa	39	5.2	1.6
South Africa	40	5.2	0.6
Bulgaria	41	4.8	1.4
Thailand	42	4.8	1.2
India	43	4.8	0.4
Greece	44	4.6	1.2
Ireland	45	4.4	1.4
Slovenia	46	4.2	1.6
Argentina	47	4	1.4
Indonesia	48	4	1
Slovakia	49	3.8	1.2
Lithuania	50	3.8	0.4
Armenia	51	3.8	0.2
Egypt	52	3.6	0.4
Mongolia	53	3.2	0.4
Dominican Republic	54	2.6	0.6
Malaysia	55	2.4	1
Tunisia	56	2.4	1
Portugal	57	2.4	0.6
Morocco	58	1.4	0.4

Let $M_{i\,2028}$ be the predicted number of medals for Country i in the 2028 Olympic Games, and $M_{i\,2024}$ be the number of medals for Country i in the 2024 Olympic Games. By taking the difference:

By taking the difference:：

$$\Delta m_{i=} M_{i\,2028_} M_{i\,2024}$$

We obtain the medal difference for Country i. A positive \Delta m indicates that the country will make progress in the medal table, while a negative \Delta m indicates that the country will regress. The final ranking results are as follows:

		Total					Total	
NOC	No	2028	2024	Δm	NOC	No	2028	2024	Δm
Cuba	1	16	9	7	Tunisia	30	2.4	3	-0.6
Germany	2	39.4	33	6.4	Morocco	31	1.4	2	-0.6
Ukraine	3	16.6	12	4.6	South Africa	32	5.2	6	-0.8
Jamaica	4	10	6	4	Brazil	33	18.8	20	-1.2
Czech Republic	5	8.8	5	3.8	India	34	4.8	6	-1.2
Kazakhstan	6	10.4	7	3.4	Thailand	35	4.8	6	-1.2
Slovakia	7	3.8	1	2.8	Norway	36	6.6	8	-1.4
Azerbaijan	8	9.4	7	2.4	Chinese Taipei	37	5.6	7	-1.4
Ethiopia	9	6.2	4	2.2	Portugal	38	2.4	4	-1.6
Mongolia	10	3.2	1	2.2	Sweden	39	8.8	11	-2.2
Colombia	11	5.8	4	1.8	Bulgaria	40	4.8	7	-2.2
Kenya	12	12.6	11	1.6	Romania	41	6.6	9	-2.4
Poland	13	11.4	10	1.4	China	42	88.4	91	-2.6
Slovenia	14	4.2	3	1.2	Ireland	43	4.4	7	-2.6
Denmark	15	10.2	9	1.2	Great Britain	44	62.4	65	-2.6
Indonesia	16	4	3	1	Hungary	45	16.4	19	-2.6
Argentina	17	4	3	1	Greece	46	4.6	8	-3.4
Serbia	18	5.8	5	0.8	Iran	47	8.4	12	-3.6
Egypt	19	3.6	3	0.6	Japan	48	41.4	45	-3.6
Malaysia	20	2.4	2	0.4	New Zealand	49	16	20	-4
Croatia	21	7.2	7	0.2	Belgium	50	5.6	10	-4.4
Mexico	22	5.2	5	0.2	South Korea	51	27.2	32	-4.8
Spain	23	18.2	18	0.2	Canada	52	22.2	27	-4.8
Georgia	24	7	7	0	Uzbekistan	53	7.6	13	-5.4
Switzerland	25	7.8	8	-0.2	Italy	54	32.6	40	-7.4
Lithuania	26	3.8	4	-0.2	Netherlands	55	25	34	-9
Armenia	27	3.8	4	-0.2	United States	56	115.2	126	-10.8
Dominican Republic	28	2.6	3	-0.4	Australia	57	41.8	53	-11.2
Turkey	29	7.4	8	-0.6	France	58	43.4	64	-20.6

The results show that the country most likely to improve is Cuba, and the country most likely to decline is France.

4.4.2Processing of non-stationary data

1. Data Selection

We use the data selected and processed in the previous part. From Group c with unstable data, we choose three representative regions/countries: Chinese Taipei, South Africa, and Uzbekistan. Then we select the medal - winning data of these three.

Note: Chinese Taipei participates in the Olympics as a Chinese regional sports team under the name of "Chinese Taipei Olympic Committee" in accordance with relevant regulations. It is an inalienable part of China's territory.

2. Data Processing

Since data from too early periods may cause excessive random fluctuations in the assessment of a country's current sports level, we only retain the competition records after 2000. Subsequently, we analyze the data, mark the athletes who have won medals multiple times (greater than or equal to twice) in the same event in the past two to three Olympic Games. We screen out these athletes as the key - focused athletes, and their corresponding sports events are the key - focused events of the target country.

3. Calculation Process

Calculate the probability of each key - focused athlete winning an award in their corresponding key - focused event.

Based on the award - winning probabilities, we quantitatively calculate the average number of medals of each athlete. The sum of the average number of medals of the key - focused athletes in this country is the predicted number of medals for this country.

$M_{t=(}\ c_{j+}\ c_{n+...)\times 3}$

M_tThe predicted medal count for country t

c_j、 c_n、 the probability of winning of country's key athletes j, n

	A	B	C
1		Probability	Total
2	Mikiko Andoh	0.166666667	0.5
3	Polina Guryeva	0.166666667	0.5
4	Hsing-chun Kuo	0.333333333	1
5	Shifang Luo	0.166666667	0.5
6	Maude Charron	0.166666667	0.5
7	Aaron Chia	0.153846154	0.461538
8	Yang Lee	0.153846154	0.461538
9	Hui Li	0.076923077	0.230769
10	Chen Liu	0.153846154	0.461538
11	Yik Soh	0.153846154	0.461538
12	Chi-lin Wang	0.153846154	0.461538
13	Keng Liang	0.076923077	0.230769
14	Chang Wang	0.076923077	0.230769
15	Total		6

Chinese Taipei

	A	B	C
1		Probability	Total
2	Lydia Jacoby	0.05	0.15
3	Lilly King	0.2	0.6
4	Annie Lazor	0.05	0.15
5	Tatjana Schoenmaker	0.1	0.3
6	Mc Mona	0.2	0.6
7	Kate Douglass	0.05	0.15
8	Qianting Tang	0.1	0.3
9	Tatjana Smith	0.1	0.3
10	Tes Schouten	0.15	0.45
11	Total		3

South Africa

	A	B	C
1		Probability	Total
2	Bekzod Abdurakhmonov	0.111111111	0.333333
3	Douglas Dake	0.222222222	0.666667
4	Mahamedkhabib Kadzimahamedau	0.222222222	0.666667
5	Zaurbek Sidakov	0.111111111	0.333333
6	Salambekovitch Jamalov	0.111111111	0.333333
7	Daichi Takatani	0.111111111	0.333333
8	Chermen Valiev	0.111111111	0.333333
9	Ulugbek Rashitov	0.181818182	0.545455
10	Hakan Recber	0.181818182	0.545455
11	Bradly Sinden	0.181818182	0.545455
12	Shuai Zhao	0.090909091	0.272727
13	Zaid Kareem	0.090909091	0.272727
14	Yushuai Liang	0.090909091	0.272727
15	Edival Pontes	0.181818182	0.545455
16	Total		6

Uzbekistan

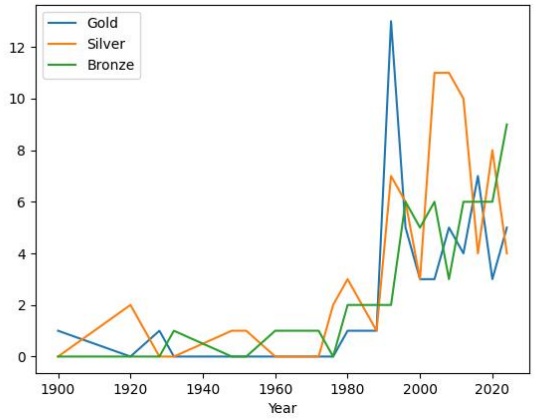


Chart of Spain 's Olmpic Medals

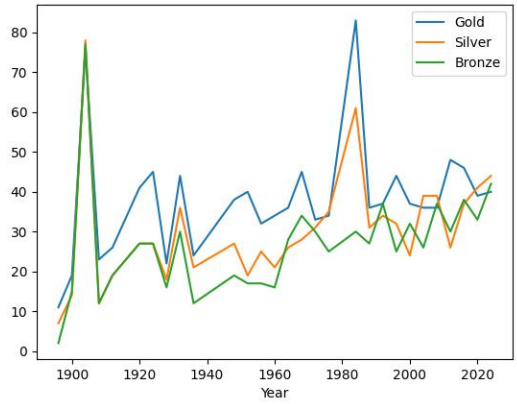


Chart of USA 's Olmpic Medals

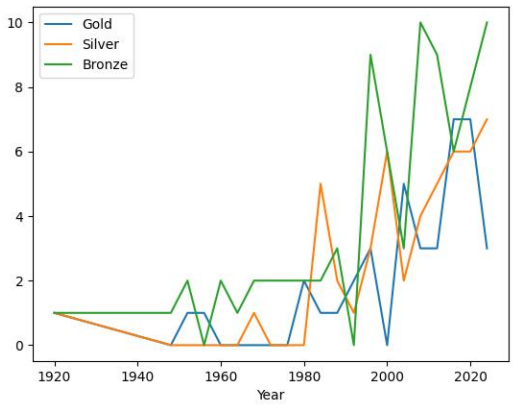


Chart of Brazil ‘s Olmpic Medals

5.Analysis of Task 2

5.1 Data screening

We identified the countries that have not won any awards through data cleaning and code - based screening..

5.2 Introduction of our method

Code1：

```
import pandas
import os
import matplotlib.pyplot as plt
import numpy as np

# Read CSV data
baseDir = './2025_Problem_C_Data'
athletes = pandas.read_csv(os.path.join(baseDir, 'summerOly_athletes.csv'), encoding='utf-8', low_memory=False)

# Filter out teams that did not participate in 2024 from athletes
athletes = athletes.groupby('NOC').filter(lambda x: x['Year'].max() >= 2024)

# Count the number of each type of medal for each NOC
medal_counts = athletes.groupby(['NOC', 'Medal']).size().unstack(fill_value=0)
medal_counts['Total'] = medal_counts['Gold'] + medal_counts['Silver'] + medal_counts['Bronze']
medal_counts = medal_counts.sort_values(['Gold', 'Silver', 'Bronze'], ascending=False)

# Filter out NOCs that have never won a medal
no_medals = medal_counts[medal_counts['Total'] == 0]
print(no_medals)

# Analyze taking ANG as an example
ang_medals = medal_counts[medal_counts.index == 'ANG']
ang_athletes = athletes[
    (athletes['NOC'] == 'ANG') &
    (athletes['Year'] >= 2020)
    ]
print(ang_athletes)

# Analyze the competition in Handball/Women Team/Handball Women's Handball events, and filter out recent Handball athletes
times = 6
handball_athletes = athletes[
    (athletes['Sport'] == 'Handball') &
    ((athletes['Event'] == 'Women') | (athletes['Event'] == 'Women Team') | (athletes['Event'] == 'Handball Women\'s Handball')) &
    (athletes['Year'] > 2024 - times * 4) &
    (athletes['Medal'] != 'No medal')
    ]

print(handball_athletes)

# Count the number of teams involved, and calculate the competition ratio (whether there is a dominant team)
team_count = handball_athletes['Team'].nunique()
print(team_count)
print(times * 3 / team_count)
```

We got the following data.

	Name	Sex	Team	NOC	Year	City	Sport	Event	Medal
222709	Paixao Af	M	Angola	ANG	2020	Tokyo	Sailing	Men's Tw	No medal
222915	Teresa Al	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
224594	Azenaide	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
224678	Magda Cc	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
225825	Wuta Dor	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
226513	Natalia F	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
227163	Salvador	M	Angola	ANG	2020	Tokyo	Swimmin	Men's 100	No medal
227354	Isabel Gu	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
228829	Albertina	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
230437	Juliana M	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
231275	Aveni Mig	M	Angola	ANG	2020	Tokyo	Athletics	Men's 100	No medal
231553	Matias Mc	M	Angola	ANG	2020	Tokyo	Sailing	Men's Tw	No medal
231672	Diassoner	F	Angola	ANG	2020	Tokyo	Judo	Women -	No medal
232696	Stelvia Pa	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
232732	Helena Pc	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
233380	Marilia Q	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
234205	Natalia Sc	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
235045	Catarina S	F	Angola	ANG	2020	Tokyo	Swimmin	Women's	No medal
235048	Helena Sc	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
236415	Liliana Vc	F	Angola	ANG	2020	Tokyo	Handball	Women T	No medal
241130	Manuela I	F	Angola	ANG	2024	Paris	Sailing	Mixed Dir	No medal
241135	Matias Mc	M	Angola	ANG	2024	Paris	Sailing	Mixed Dir	No medal
242417	Manuel A	M	Angola	ANG	2024	Paris	Canoe Sp	Men's Car	No medal
242420	Benilson S	M	Angola	ANG	2024	Paris	Canoe Sp	Men's Car	No medal
242421	Benilson S	M	Angola	ANG	2024	Paris	Canoe Sp	Men's Car	No medal
242422	Andre Mc	M	Angola	ANG	2024	Paris	Rowing	Men's Sin	No medal
243665	Lia Lima	F	Angola	ANG	2024	Paris	Swimmin	Women's	No medal
243667	Henrique	M	Angola	ANG	2024	Paris	Swimmin	Men's 100	No medal
243686	Marta Alt	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243709	Eliane Pa	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243710	Marilia Q	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243712	Chelcia G	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243714	Dolores R	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243718	Liliane M	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243769	Vilma Ne	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243776	Juliana M	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243865	Natalia F	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243868	Azenaide	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243872	Stelvia Pa	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243879	Albertina	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243882	Helena Pc	F	Angola	ANG	2024	Paris	Handball	Women	No medal
243885	Natalia Sc	F	Angola	ANG	2024	Paris	Handball	Women	No medal
244899	Filipe An	M	Angola	ANG	2024	Paris	Sailing	Men's Dir	No medal
246711	Teresa Al	F	Angola	ANG	2024	Paris	Handball	Women	No medal
247247	Marcos Sc	M	Angola	ANG	2024	Paris	Athletics	Men's 100	No medal
250629	Edmilson	M	Angola	ANG	2024	Paris	Judo	Men -66 k	No medal

	Name	Sex	Team	NOC	Year	City	Sport	Event	Medal
33	Ragnhild	F	Norway	NOR	2008	Beijing	Handball	Handball	Gold
1966	Macarena	F	Spain	ESP	2012	London	Handball	Handball	Bronze
3631	Nely Fran	F	Spain	ESP	2012	London	Handball	Handball	Bronze
4734	Jessica Be	F	Spain	ESP	2012	London	Handball	Handball	Bronze
4821	Ida Alstac	F	Norway	NOR	2012	London	Handball	Handball	Gold
4822	Ida Alstac	F	Norway	NOR	2016	Rio de Jar	Handball	Handball	Bronze
5507	Vanessa C	F	Spain	ESP	2012	London	Handball	Handball	Bronze
5644	An Jeong-	F	South Kor	KOR	2008	Beijing	Handball	Handball	Bronze
5890	Kristine A	F	Denmark	DEN	2004	Athina	Handball	Handball	Gold
8185	Emilie Ar	F	Norway	NOR	2016	Rio de Jar	Handball	Handball	Bronze
9762	Camille A	F	France	FRA	2016	Rio de Jar	Handball	Handball	Silver
10546	Bae Min-k	F	South Kor	KOR	2008	Beijing	Handball	Handball	Bronze
12422	Sonja Bar	F	Monteneg	MNE	2012	London	Handball	Handball	Silver
12602	Andrea M	F	Spain	ESP	2012	London	Handball	Handball	Bronze
21729	Anastasiy	F	Ukraine	UKR	2004	Athina	Handball	Handball	Bronze
21911	Nataliya I	F	Ukraine	UKR	2004	Athina	Handball	Handball	Bronze
23975	Karoline I	F	Norway	NOR	2008	Beijing	Handball	Handball	Gold
23976	Karoline I	F	Norway	NOR	2012	London	Handball	Handball	Gold
25772	Karen Brc	F	Denmark	DEN	2004	Athina	Handball	Handball	Gold
26436	Anela Bul	F	Monteneg	MNE	2012	London	Handball	Handball	Silver
26438	Katarina	F	Monteneg	MNE	2012	London	Handball	Handball	Silver
26509	Chlo Bull	F	France	FRA	2016	Rio de Jar	Handball	Handball	Silver
26908	Hanna Bu	F	Ukraine	UKR	2004	Athina	Handball	Handball	Bronze
34176	Choi Im-je	F	South Kor	KOR	2004	Athina	Handball	Handball	Silver

This figure shows the events that ANG has signed up for since 2020. From this, we can predict the events that the country intends to focus on developing, or the events in which the key - trained athletes participate. Therefore, we use Code 1 to find Figure 2, which shows the main competing teams in this event over four Olympic Games. We call a function to calculate the competition ratio of each competing team to check if there is a team that always wins, so as to observe the medal - winning competitiveness of ANG in this event.

```python
import numpy as np
import pandas as pd
import os
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

# Read data
baseDir = './2025_Problem_C_Data'
athletes = pd.read_csv(os.path.join(baseDir, 'summerOly_athletes.csv'), encoding='utf-8', low_memory=False)

# Define a function to calculate the number of medals
def count_medals(df):
    # Group by 'NOC', 'Year', 'Medal' and count the number of medals, then unstack the result
    medal_counts = df.groupby(['NOC', 'Year', 'Medal']).size().unstack(fill_value=0)
    # Calculate the total number of medals
    medal_counts['Total'] = medal_counts['Gold'] + medal_counts['Silver'] + medal_counts['Bronze']
    # Reset the index
    medal_counts = medal_counts.reset_index()
    return medal_counts

# Calculate the number of medals for each country each year
medal_counts = count_medals(athletes)

# Filter out the countries that have not won any medals
countries_no_medal = medal_counts.groupby('NOC')['Total'].sum()
countries_no_medal = countries_no_medal[countries_no_medal == 0].index

# Filter out the historical data of these countries
no_medal_data = medal_counts[medal_counts['NOC'].isin(countries_no_medal)].copy()

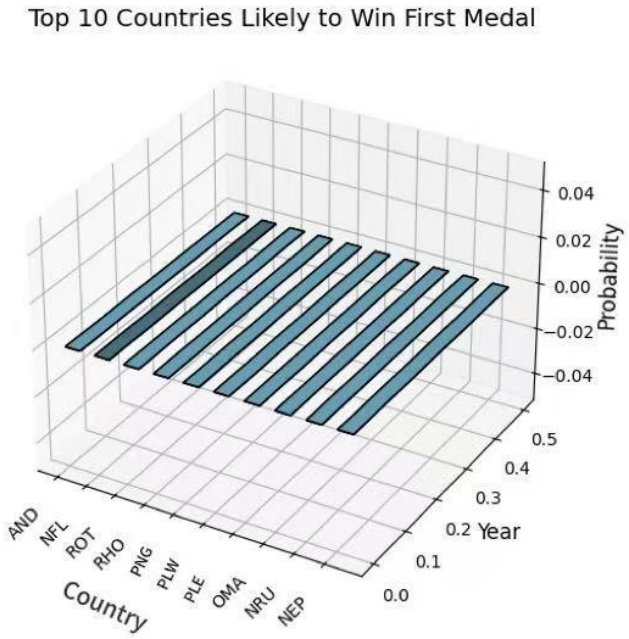
# Calculate the medal trends of these countries over the years
no_medal_data.loc[:, 'trend'] = no_medal_data.groupby('NOC')['Total'].diff().fillna(0)

# Calculate the probability of these countries winning their first medal
medal_probabilities = no_medal_data.groupby('NOC')['trend'].apply(lambda x: np.mean(x > 0))
# Clip the probability values between 0 and 1
medal_probabilities = medal_probabilities.clip(0, 1)

# Sort the probability values
medal_probabilities = medal_probabilities.sort_values(ascending=False)

# Select the top 10 countries
top_countries = medal_probabilities.head(10)
```

We use Code 2 to calculate the medal - winning trends of various countries, thereby calculating the medal - winning probabilities of countries that haven't won any medals yet. Then we obtain the rankings, which are shown in the following bar chart.



According to the bar chart, the probabilities of countries that have not won medals yet to win their first medal are similar, with AND ranking first. Following it are NFL, ROT, and so on.

6.Analysis of Task 3

6.1 Data processing

a. Take the event category as the antecedent, and sort out which countries have a high proportion of medal counts in the same event.
b. Take the country as the antecedent, and sort out which event accounts for the highest proportion of a certain country's medal count.

6.2 Introduction of method

The Apriori algorithm we referenced is a classic association rule learning algorithm, commonly used to mine frequent itemsets and their association rules in large-scale datasets. Its working principle is to screen out frequent itemsets based on a support threshold, generate association rules from these frequent itemsets, and then evaluate the effectiveness of the rules by calculating the confidence level. After evaluation, this algorithm is suitable for finding the advantageous sports events of different countries.

6.3Process

To solve the task of the most important sports events for different countries:
a. Apply the Apriori algorithm to generate association rules between countries and events.
b. Calculate the confidence coefficient
c. The final results are divided into two tables according to taking the country as the antecedent and taking the event as the antecedent. Sort them in descending order of the confidence level. The higher the confidence level, the more obvious the advantage.

antecedent	consequent	confidence coefficient
FIJ	Rugby Sevens	1
MAR	Football	0.95
ARG	Hockey	0.842105263
ROU	Rowing	0.80952381
DEN	Handball	0.756097561
IND	Hockey	0.727272727
NOR	Handball	0.652173913
CRO	Water Polo	0.65
RSA	Rugby Sevens	0.619047619
POL	Volleyball	0.464285714
SRB	Water Polo	0.448275862
HUN	Canoe Sprint	0.433333333
AUS	Swimming	0.421487603
SRB	Basketball	0.413793103
BRA	Football	0.328358209
NED	Hockey	0.296610169
ESP	Football	0.265060241
KOR	Judo	0.25862069
GBR	Athletics	0.256410256
JPN	Judo	0.256097561
CAN	Rugby Sevens	0.24561403
NZL	Cycling Track	0.235294118
NZL	Rugby Sevens	0.235294118
GBR	Rowing	0.230769231
NED	Rowing	0.228813559
USA	Swimming	0.214953271
CHN	Swimming	0.214285714
JPN	Fencing	0.207317073
KOR	Archery	0.206896552
Judo	JPN	0.2
BRA	Volleyball	0.194029851
BRA	Judo	0.194029851
GER	Football	0.185185185
ESP	Handball	0.180722892
USA	Athletics	0.180685358
ITA	Fencing	0.170731707
GER	Hockey	0.166666667
ITA	Volleyball	0.158536585
ESP	Water Polo	0.156626506
GBR	Cycling Track	0.153846154
GER	Handball	0.148148148
NED	Athletics	0.144067797
Judo	KOR	0.142857143
FRA	Basketball	0.128342246
FRA	Judo	0.122994652
FRA	Football	0.112299465
GER	Canoe Sprint	0.111111111
NED	Water Polo	0.110169492
AUS	Water Polo	0.107438017
CHN	Hockey	0.101190476

antecedent	consequen	confidence coefficien
Diving	CHN	0.41666667
Archery	KOR	0.4
Hockey	NED	0.343137255
Basketball	USA	0.338028169
Basketball	FRA	0.338028169
Volleyball	USA	0.333333333
Handball	DEN	0.329787234
Swimming	USA	0.322429907
Equestrian	GBR	0.311111111
Cycling Track	GBR	0.296296296
Artistic Gymnastics	USA	0.253731343
Athletics	USA	0.252173913
Rowing	GBR	0.25
Swimming	AUS	0.238317757
Shooting	CHN	0.222222222
Judo	FRA	0.219047619
Canoe Sprint	HUN	0.209677419
Judo	JPN	0.2
Artistic Gymnastics	CHN	0.194029851
Canoe Sprint	GER	0.193548387
Fencing	JPN	0.188888889
Rowing	NED	0.1875
Handball	FRA	0.180851064
Rugby Sevens	CAN	0.179487179
Rugby Sevens	FIJ	0.179487179
Fencing	FRA	0.177777778
Football	BRA	0.177419355
Football	ESP	0.177419355
Hockey	GER	0.176470588
Athletics	GBR	0.173913043
Handball	GER	0.170212766
Football	FRA	0.169354839
Basketball	SRB	0.169014085
Swimming	CHN	0.168224299
Water Polo	NED	0.166666667
Volleyball	BRA	0.166666667
Water Polo	SRB	0.166666667
Water Polo	ESP	0.166666667
Water Polo	AUS	0.166666667
Hockey	CHN	0.166666667
Volleyball	ITA	0.166666667
Water Polo	CRO	0.166666667
Water Polo	USA	0.166666667
Volleyball	POL	0.166666667
Rugby Sevens	FRA	0.166666667
Rugby Sevens	RSA	0.166666667
Volleyball	FRA	0.166666667
Football	USA	0.161290323
Football	GER	0.161290323
Handball	ESP	0.159574468
Handball	NOR	0.159574468
Hockey	IND	0.156862745
Hockey	ARG	0.156862745
Fencing	ITA	0.155555556
Rugby Sevens	USA	0.153846154
Rugby Sevens	NZL	0.153846154
Football	MAR	0.153225806
Cycling Track	NZL	0.148148148
Cycling Track	AUS	0.148148148
Judo	KOR	0.142857143
Judo	BRA	0.123809524
Rowing	ROU	0.118055556

To explore the Impact of the Event Selection by the Host Country on the Medal Distribution

a. Based on the final tabular data in 1.3.1, we can filter the data with the antecedent being USA (the host country of the 2028 Olympic Games). It shows that the USA has advantages in the major event categories of Swimming and Athletics.
b. Filter the data with the antecedents being Swimming and Athletics. It shows that in these two major event categories, the USA, AUS, GBR, and CHN have advantages.
c. We also organized the medal ranking data of the host countries in past Olympic Games. From the charts, we can see that being the host country has an advantage in increasing the number of medals of the host country in that Olympic Games.
d. Besides the fact that the host country's athletes are familiar with the local climate and venues, which may affect the athletes' performance, the main reason for the increase in the number of medals of the host country is that the host country usually adds awards in the major event categories where it has advantages. Therefore, we can predict that in the 2028 Olympic Games, the number of medals of AUS, GBR, and CHN, which have the same advantageous event categories as the USA, may increase.

6.4 Result

antecedent	consequent	confidence coefficient
USA	Swimming	0.214953271
USA	Athletics	0.180685358

antecedent	consequent	confidence coefficient
Swimming	USA	0.322429907
Athletics	USA	0.252173913
Swimming	AUS	0.238317757
Athletics	GBR	0.173913043
Swimming	CHN	0.168224299

7.Analysis of Task 4

7.1 The precondition of investing in "great coaches“

a. The country has won honors in this sport before, which indicates that the country has the ability to win awards in the same sport in terms of ethnicity, sports facilities and the strength of athletes.
b. The country's performance declined significantly due to the departure of a certain coach.
c. The country possesses a certain economic strength and has capacity to hire coaches.

7.2 Three Example of Country

Country：India

Sport：Football

Historical score：In 1951, India had won the gold medal in football at the Asian Games. The coach of the team was Abdul Rahim, who was famous in the world. But he lost control of the team the following year before the Olympic Games.Finally,the team lost 1:10 to Yugoslavia. After the death of Abdul Rahim in 1963, Indian football lost the guidance of professional coaches, and its performance is still unsatisfactory today. [1]

Reason analysis：In the field of football, India once won the first place in Asia, which indicates that it has the ability to win medals in terms of both the human race and the strength of athletes. However, due to the sudden departure of the coach, the strength of the team dropped sharply, which is sufficient to show that the influence of the coach on the team is of vital importance, and this is what India is lacking at present. In addition, according to the latest data, India's total GDP ranked fifth in the world in 2023, which means that India has the economic strength to invest in "great coach".[2]

Country：Katar

Sport：Football

Historical score：On December 11, 2024, the Qatar Football Association announced that the national team's head coach, Max Lopez, had resigned for the recent poor performance. In the

[1] https://www.sohu.com/a/519016202_121119642

[2] https://web.phb123.com/city/GDP/view_IN.html?show_loading=0&webview_progress_bar=1

2024 World Cup qualifiers Asian 18-team playoffs, Qatar ranked fourth in Group A which meant that it had a poor performance among countries of the same race.[3]

Reason analysis： Qatar has a lot of room for improvement in football personnel selection and tactical distribution.All these fall within the responsibility of the coach. Therefore, the team urgently needs an outstanding coach to lead. Moreover， Qatar attaches great importance to the construction of sports youth training system, especially in the innovation of football concept and technology. Besides, Qatar is rich in oil and natural gas resources so that it has strong financial power to provide a strong financial guarantee for sports development to invest in "great coaches".

3

Country : Brazil

Sport: Tennis

History: Brazil had won MEDALS in the 2020 Olympics[4] but won no MEDALS in tennis at the 2024 Olympics. 。[5]

Reason analysis: Brazil ,who is always in the top of the Olympic medal table. But it never got good score in tennis. A famous coach can strengthen the construction of Brazilian tennis team, improve the level of Brazilian tennis, so that Brazil's results in the medal table might be well in the upcoming Olympics.

8. Analysis of Task 5

8.1 Insights about Olympic medal counts

1. Stability of Medal Trends

Through the stationarity test, if a country's medal data is stationary, it indicates that the country's ability to obtain medals in the Olympics is relatively stable. This may be due to a mature sports talent cultivation system, stable sports investment, or long - term accumulated advantages in specific sports. Conversely, if the data is non - stationary, it means that the country's medal - obtaining situation fluctuates greatly, possibly affected by various unstable factors such as domestic sports policy adjustments, sudden emergence or loss of sports talents.

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[3] https://www.sohu.com/a/835389570_122066679

[4] https://m.163.com/dy/article/J99HOKFH0549O2UA.html?webview_progress_bar=1&show_loading=0&referFrom=

[5] https://weibo.com/ttarticle/p/show?id=23094050640488312344 53

2. Potential Laws of Medal Growth or Decline

The parameters of the ARIMA model (such as autoregressive terms, differencing orders, and moving average terms) can reflect the potential laws of medal count changes. The autoregressive terms show the degree of influence of past medal counts on the future; the differencing order can indicate whether the change trend of medal data is linear or non - linear; the moving average terms reflect the influence of random interference factors on medal counts.

For instance, if the autoregressive coefficient is large, it means that past medal achievements have a significant positive impact on the future, and the country's advantages in sports events are continuous. If the differencing order is high, it may mean that the change in the country's medal count is not a simple linear increase or decrease but is affected by the interaction of various complex factors.

8.2 Suggestions for National Olympic Committees:

1. Formulating Long - Term Sports Development Strategies
For countries with stationary medal data, the NOC can continue to consolidate the existing sports development model, increase investment in advantageous sports, and maintain a leading position. For countries with non - stationary medal data, the NOC needs to conduct in - depth analysis of the reasons for the fluctuations, adjust sports policies, and optimize the talent cultivation system to improve the stability of medal acquisition.

2. Based on the laws of medal growth or decline revealed by the ARIMA model, and based on frequent item sets and association rules mined by Apriori algorithm, the NOC can allocate resources reasonably. For sports with growth potential, it can increase investment in funds, human resources, and materials, including building training facilities and hiring professional coaches which’s power have been showed by lots of examples mentioned above. For sports with limited development or a downward trend, it can appropriately reduce resource investment or carry out sports transformation.

9. Conclusion

In conclusion, this paper uses the multiple mathematical models and algorithms to provide a solution for solving the prediction of Olympic medal table. And through logical reasoning, it provides helpful experience for the Olympic Committee.Also the current models for predicting medal counts can be further refined by incorporating more complex variables such as athletes condition in different sports competition season.For the effect of great coach, we can establish a database of the famous coach to match the team.In addition, we hope that the predicted model can be applied in wider field.

Reference

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Report on use of AI

1. We used ai to translate the some sentences and polish the unidiomatic expressions during the paper - writing process. All the language polished by AI was strictly reviewed by the team before being applied to the corresponding parts of the paper. All AI - generated texts were produced with rigorous prompts and sufficient content, and the team did not abuse AI.

2.We use ai to improve code quality.